A System Design Document of Academic Thesis Management and Documentation System

COSC 75 : Software Engineering II

Instructor
Ms. Cyril Oliva

Prepared by
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System Architecture

This system architecture shows how the Academic Thesis Management and Documentation System is built and how its parts work together. It presents the connection between the front-end, back-end, and database components that support core functions like thesis submission, review, and storage. Through this structure, the system ensures smooth operation and easy access to thesis-related information.

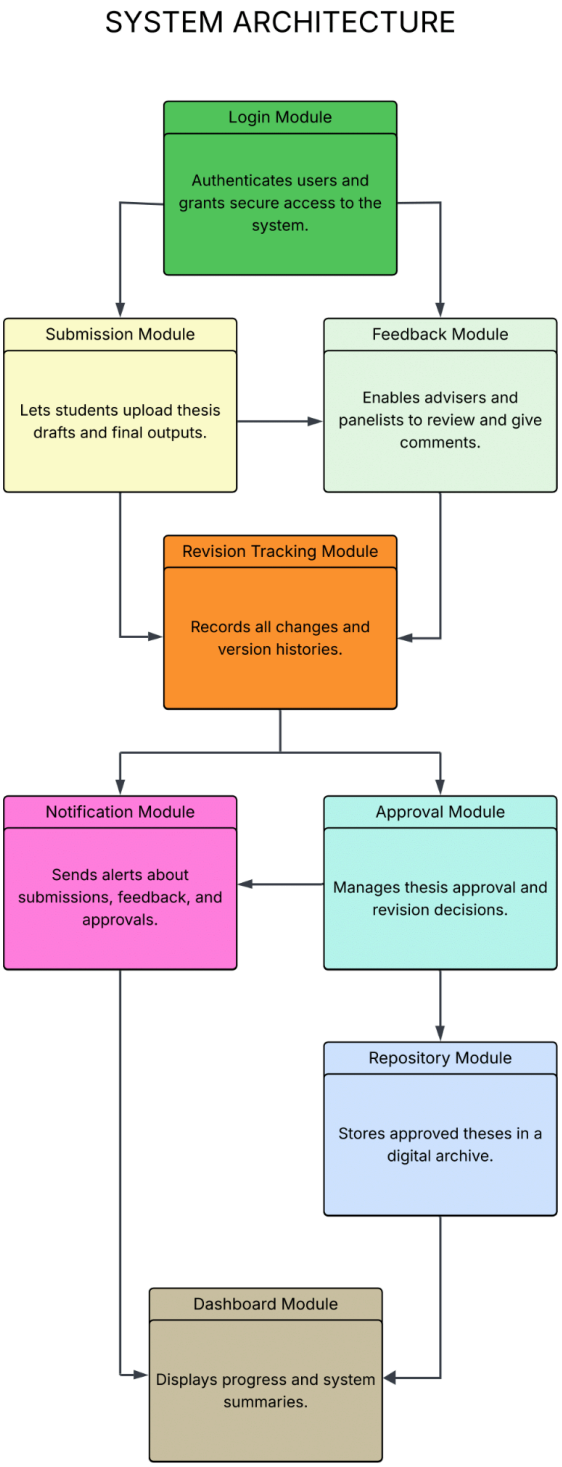


Figure 1.

Figure 1. System Architecture (Functional Modules)

The diagram above illustrates the overall structure and interaction between the system's main modules. It begins with the **Login Module**, which provides secure user authentication. Once logged in, users can upload their thesis drafts through the **Submission Module**, while advisers and panelists review them via the **Feedback Module**. All changes and version histories are recorded on the Revision Tracking Module. The Notification Module sends alerts regarding submissions, feedback, and approvals, while the **Approval Module** handles thesis evaluation and decision-making. Approved theses are then stored in the **Repository Module**, and overall progress and summaries are displayed through the **Dashboard Module**.

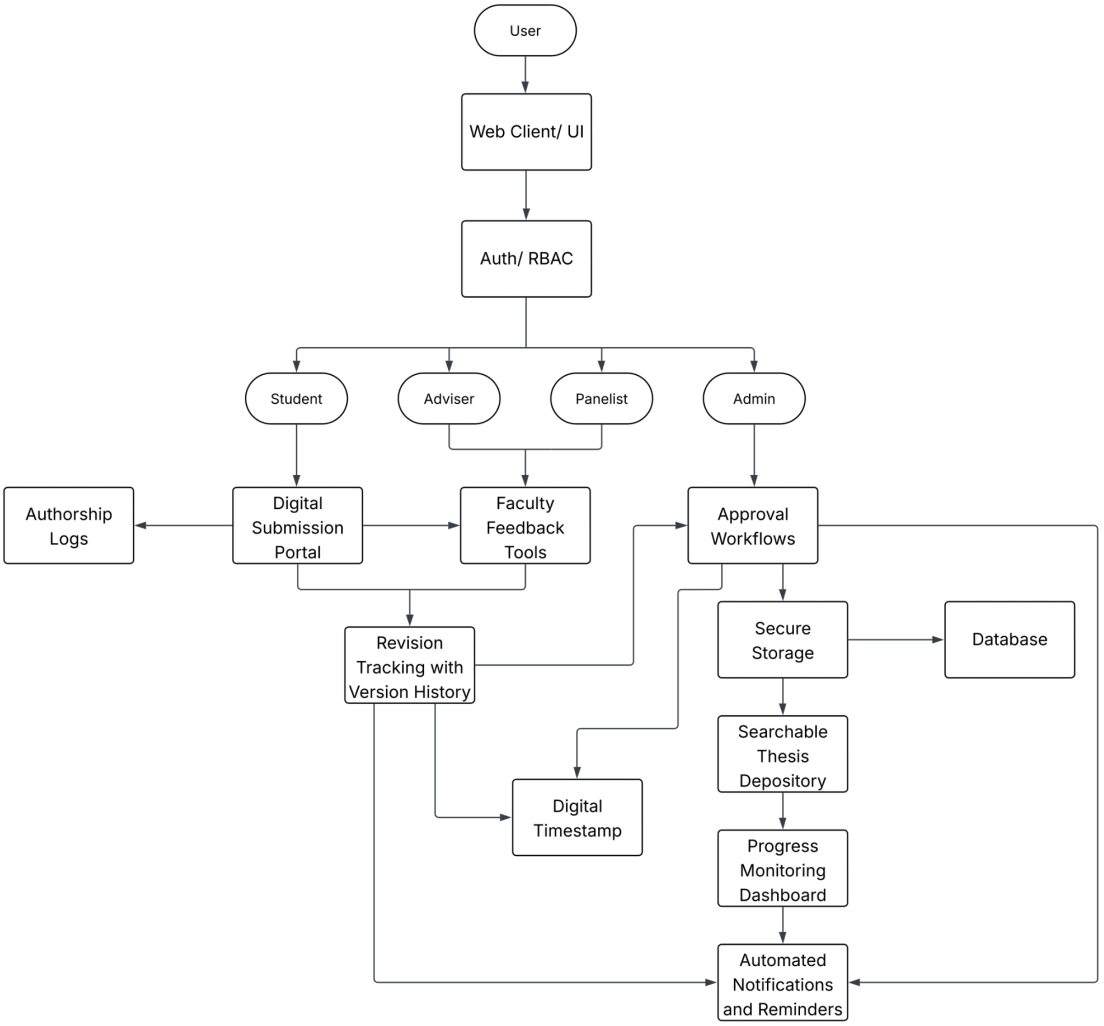


Figure 2. Layered System Architecture

Unlike Figure 1, which focuses on the system's functional modules and the flow of operations between them, Figure 2 presents a more detailed view of the **system's layered structure**. It illustrates how different users: **Students, Advisers, Panelists, and Administrators** interact with the system through the **Web Client/UI** and are authenticated using **Role-Based Access Control (RBAC)**. Each user role is connected to specific components such as the Digital **Submission Portal**, **Faculty Feedback Tools**, and **Approval Workflows**, depending on their permissions. The Diagram also highlights backend processes including **Revision Tracking with Version History**, **Secure Storage**,

and a **Searchable Thesis Depository** integrated with the **Database**. Supporting features like **Digital Timestamping**, **Monitoring Dashboard**, and **Automated Notifications** ensure accountability, transparency, and seamless coordination throughout the thesis management process.

Figure 3. Technical Architecture

This figure focuses on the **technical structure** of the system, illustrating how different technologies and components interact. It highlights the connection between the frontend, backend, database, repository, and notification system, showing how each part works together to support the system's overall functionality and performance.

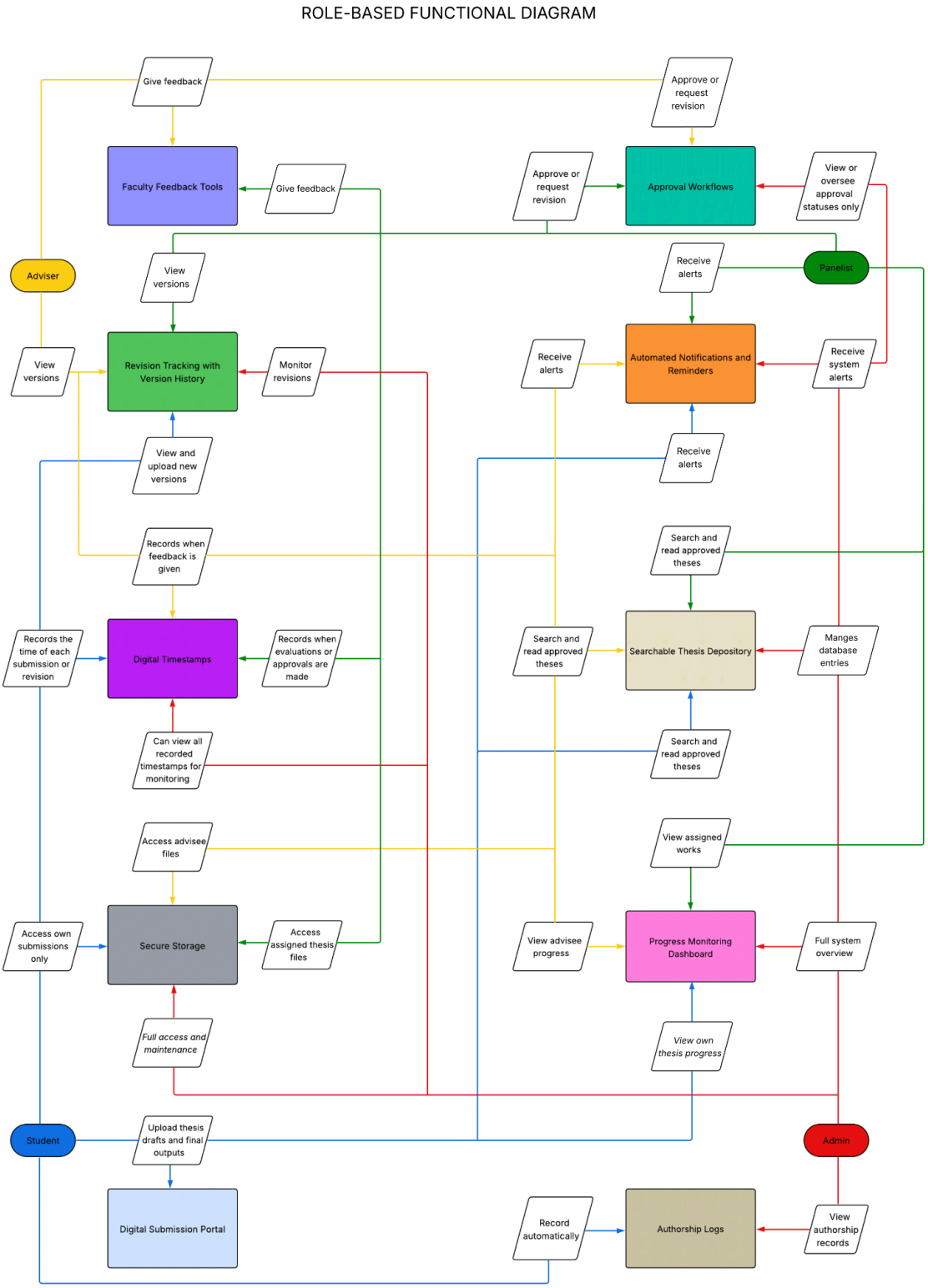
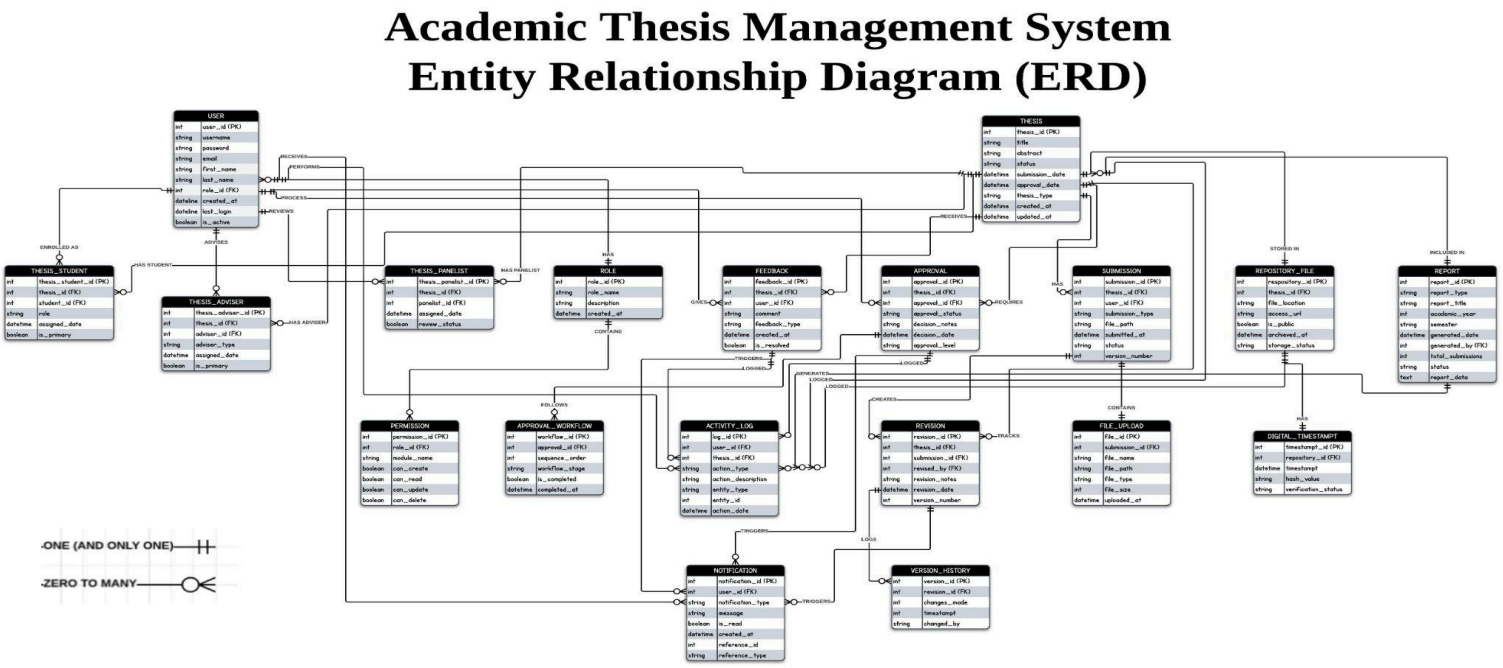


Figure 4. Role-Based Functional Diagram

This figure presents the role-based functional architecture of the system. It shows the distinct responsibilities and system interactions of each user type in performing key processes. The diagram emphasizes how user roles are integrated across various system modules to promote collaboration, transparency, and efficient thesis documentation management.

Database Design

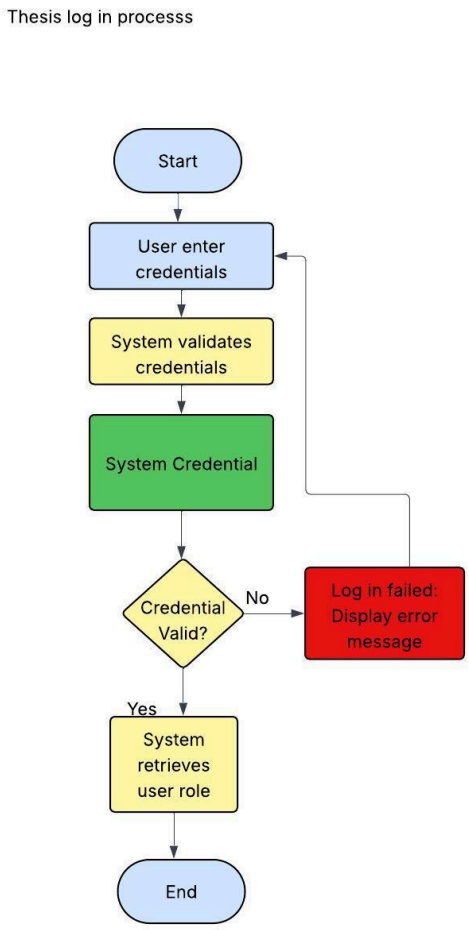
The database design defines the structure and organization of data within the Academic Thesis Management and Documentation System. It ensures that all thesis records, user information, feedback, revisions, and approvals are efficiently stored, retrieved, and maintained. This design supports the system's processes by providing a reliable and consistent foundation for data management and integrity.

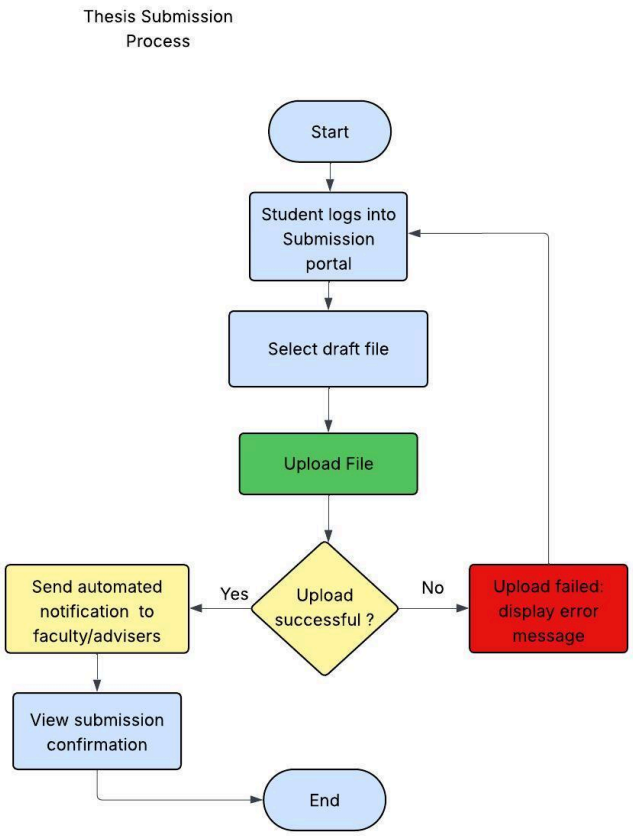


[For the clearer view/copy of the ERD]

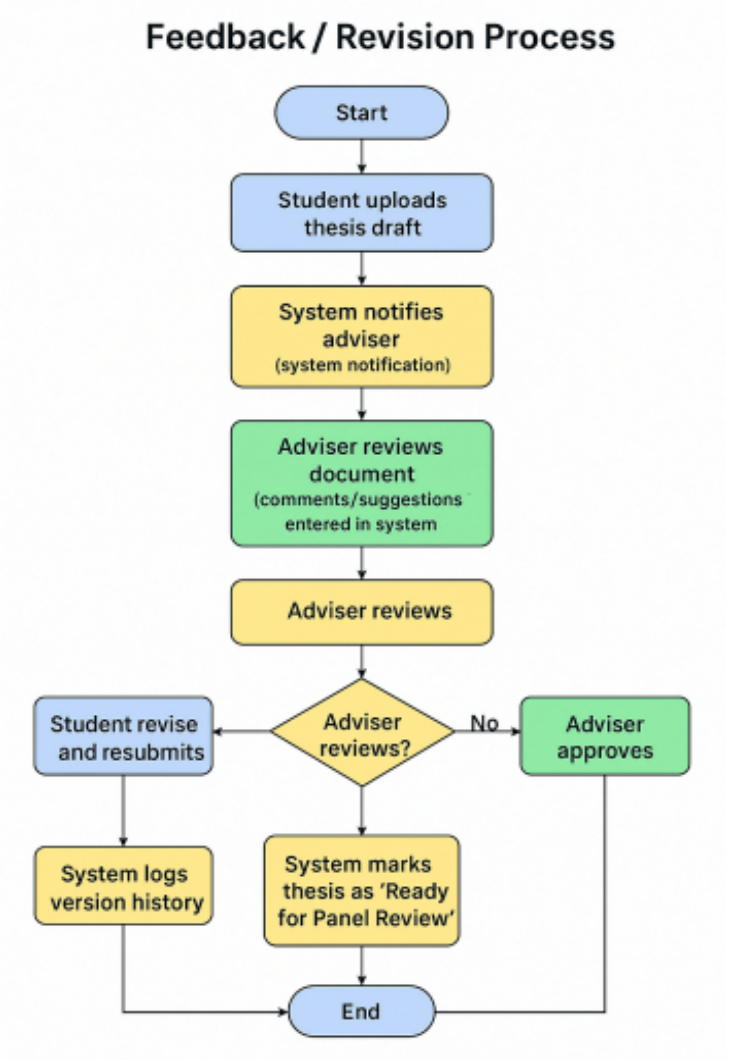
Process Design

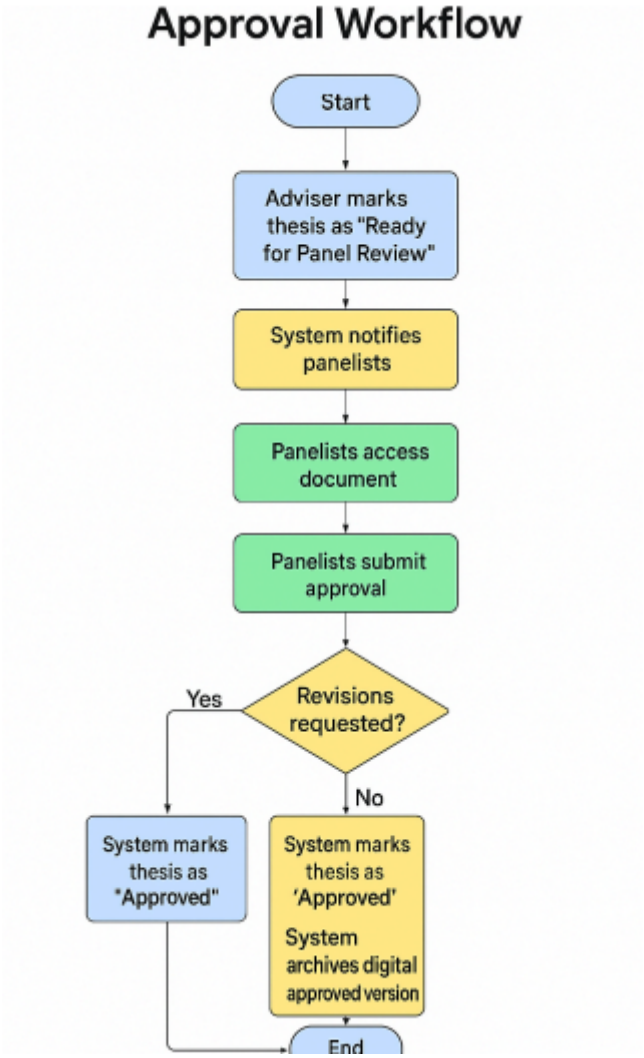
The process design presents the systematic flow of operations within the Academic Thesis Management and Documentation System. It defines how each module interacts to perform specific functions such as user authentication, thesis submission, feedback management, approval, notification, and archiving. The design focuses on ensuring logical sequencing, data consistency, and smooth coordination among all processes. By outlining the interaction between system components, the process design serves as a blueprint for implementing an efficient and automated thesis management workflow.



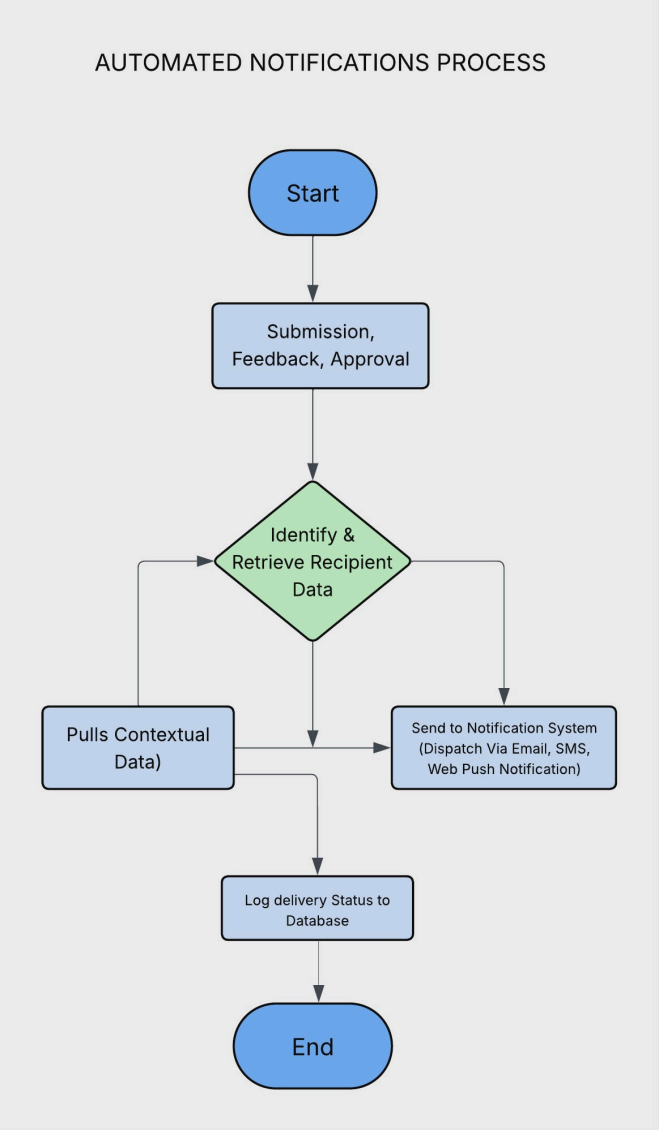


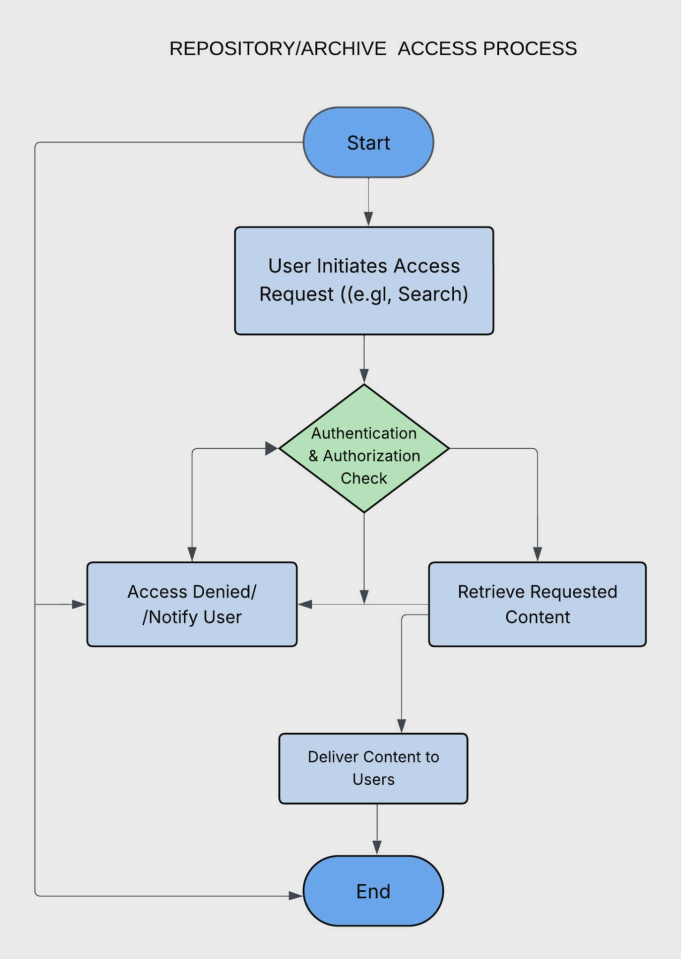
The process begins when a user enters their login credentials into the system. The system validates the input and determines if access is granted based on the user's role. Invalid credentials prompt an error message, allowing the user to retry or reset their password, while successful authentication logs the activity for version tracking. Once logged in, the student proceeds to the thesis submission portal to upload their draft file. The system verifies the upload and confirms if it is successful. In case of failure, the user is prompted to re-upload the file. Upon successful submission, the system automatically sends notifications to the adviser or faculty and displays a confirmation message to the student, ensuring a smooth and transparent submission workflow.





The feedback and Revision Process begins when a student uploads their thesis draft to the system. The Adviser is automatically notified and can access the document for review. Feedback is entered directly into the system, allowing the student to view comments and suggestions. The student then makes necessary revisions and reuploads the updated file. The system maintains version history for transparency and tracking. This cycle continues until the adviser confirms that the thesis meets all academic standards and can move to the Approval Workflow.





The process begins when actions such as submission, feedback, or approval occur within the system. The system automatically identifies and retrieves recipient data, then sends notifications through various channels such as email, SMS, or web push alerts. Delivery statuses are logged in the database to ensure transparency and tracking.

Meanwhile, the repository or archive access process starts when a user initiates an access request, such as searching for a document. The system performs authentication and authorization checks to verify user permissions. If the request is valid, the requested content is retrieved and delivered to the user; otherwise, access is denied and the user is notified. Together, these processes ensure seamless communication and secure access to stored academic resources.